

12 Builder Basics · Variables — English Worksheet (Intermediate)

Topic: Variables · **Course:** Builder Basics · **Level:** Intermediate · **Time:** about 38 minutes

1 · Predict 🧠

A **variable** is a name that holds a number. `length = 5` means the name `length` now holds 5.

Read each set of steps. Write what you think will happen.

```
length = 4
repeat length times:
  place block down
  move forward
```

What does the agent build? How long is it?


```
length = 8
repeat length times:
  place block down
  move forward
```

Now the first line changes to `length = 3`. How long is the path? How many lines did you have to edit?

2 · Find the Bug 🐛

Each code block is broken. Write one or two sentences on what is wrong.

Bug A — The agent should build a path 6 blocks long.

```
repeat length times:
  place block down
  move forward
length = 6
```

Hint: think about the order — when does the name get its number?

What is wrong?

Bug B — One variable should control both sides of the corner.

```
length = 6
repeat length times:
  place block down
  move forward
turn right
repeat 4 times:
  place block down
  move forward
```

What is wrong, and which part would you change?

3 · Fix the Code

The agent should build a path **7 blocks long** using a variable. The code has two problems.

```
lenght = 7
repeat length times:
  move forward
  place block down
```

Write the fixed code:

Why does your version work? Write one sentence.

4 · Show Your Work 📷 🎥

Switch to your homework world. Build a **path or fence** using a variable called `length`. Run it. Then change the value of `length` **one time** and run it again, so you get a second build of a different size.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what each part does. Point at the line that sets your variable.

2 · Your build. Point the camera. Show both sizes you built.

Fill the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

The most fun part was _____.

Something new I learned was _____.

Write your lines here, then say them in your video.

Submit ✓

Send this worksheet + your video to teacher on KakaoTalk.